Research Paper Summary

Use of Network Pharmacology and Molecular Docking to Explore the Mechanism of Action of Curcuma in the Treatment of Osteosarcoma

Osteosarcoma (OS) is a malignant bone tumor originating from bone tissue. The standard treatment for OS includes surgical resection combined with preoperative and postoperative adjuvant radiotherapies. However, the metastasis, invasion, and recurrence rates of OS remain high and the use of excessive chemotherapy drugs increases the risk of adverse reactions and aggravates tumor resistance. Therefore, a safe and effective drug should be developed to help reduce OS cell invasion and increase its chemosensitivity. One candidate would be Curcuma which contains compounds with anticancer activity. Curcumin in particular inhibits the proliferation of OS cells and induces their apoptosis. The active antitumor compounds in Curcuma for OS treatment require further study. **(Research Background → 109 words)**

This study is aimed to determine the therapeutic targets and potential signaling pathways most closely related to OS to determine the potential mechanism of Curcuma as an anticancer agent in this disease. **(Research Objectives → 32 words)**

The study starts by obtaining and screening using the Swiss ADME database based GI absorption and drug likeness. Databases of OS-associated targets were found for identification. The STRING database was used to analyze the relationship of common Curcuma and OS-related targets for predicting protein–protein interactions. All potential therapeutic targets were subjected to Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis. The results are shown as a graph of signaling pathways and a bubble plot of GO categories. Molecular docking is used to analyze the relationship between the key targets of curcuma and the core compound. **(Research Workflow → 101 words)**

A total of 270 active components and 3688 corresponding targets were identified, with 141 targets identified and considered potential therapeutic targets of curcuma used in the treatment of OS. From the PPI network, 12 hub genes were identified: AKT1, TNF, STAT3, EGFR, HSP90AA1, ESR1, EP300, ERBB2, PTGS2, MDM2, TLR4, and AR. The molecular docking results showed that curcumin had a strong affinity for AKT1, TNF, STAT3, and EGFR, with low RMSD values indicating reliable results. **(Key Results → 75 words)**

Overall, the results based on network pharmacology and molecular docking presented in this study provide important insights into the potential anticancer mechanisms of curcuma by highlighting its ability to interact with multiple targets involved in cancer development and progression. Despite these findings, there are still some limitations to this study, including the development of disease that involves a pathological process, the curative effect of the compound differs significantly at each stage, and the therapeutic effects of different doses of curcuma on the treatment of OS should be considered. Additionally, since the study was based on database analysis and computational approaches, there is a lack of experimental findings to validate the results of our research. **(Strengths and Limitations → 114 words)**

In this study, network pharmacology and molecular docking revealed that curcuma-mediated treatment of OS was a complex process. Through signaling pathways, Curcuma regulates key targets that are involved in the treatment of OS. These targets and pathways provide a pharmacological basis for the clinical treatment and a theoretical basis for the development of new drugs that interfere with lung metastasis and drug resistance in OS. (Key Insights → 65 words)

Total=496 words

Methods and Results Illustration